Main fuel system components - the layout and primary components are identical for 53, 71 and 92 Series engines.

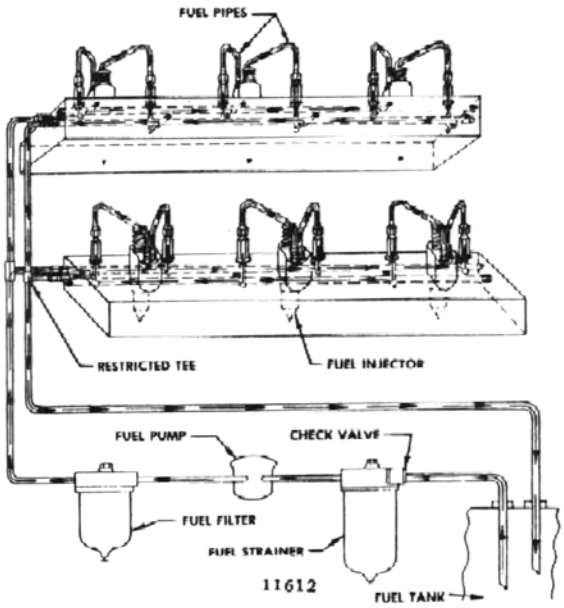


Fig. 1 – Schematic Diagram of Typical Fuel System

Fuel filters

Primary - located in series between the fuel tank and the engine mounted gear pump, this filter is under suction. Older applications used non-disposable canisters within which was a disposable element. Fitted with an inlet check valve on the mounting pad.

Secondary - located in series between the gear pump and the inlet port in the cylinder head fuel manifold. Current secondary filters are disposable, spin-on filter cartridges, which entrap particles down to 1 micron in size and may plug on water.

Gear pump

An engine driven, positive displacement pump responsible for charging the unit injectors with pressures between 30-70 psi (200 470 KPa). Also called a transfer pump.

Fuel lines and fuel manifold

Hydraulic hose is used to connect the fuel tank with the filters, pump, and deliver fuel to the fuel manifolds. In 53, 71 and 92 Series, the fuel manifolds are drillings within the cylinder head(s) that deliver fuel to the unit injectors at charging pressure and return fuel back to the tank(s). A restriction fitting is located at the exit port of each return fuel manifold. This defines the flow area that establishes the charging pressure window. A relief valve in the transfer pump prevents the charging pressure from exceeding 70 psi.

Jumper pipes

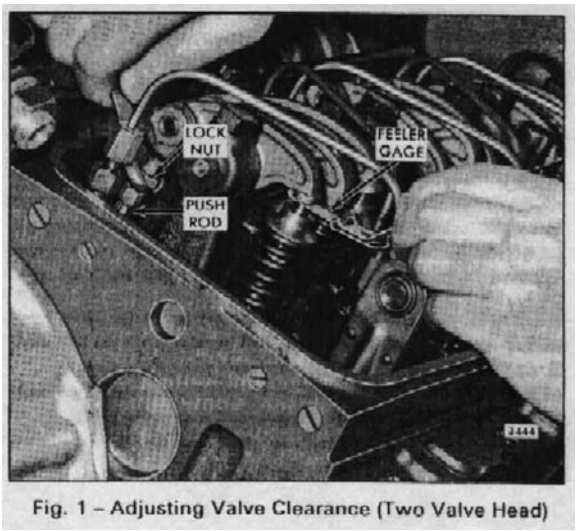
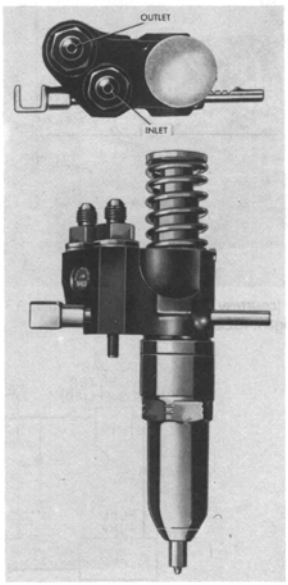


Fig. 1 – Adjusting Valve Clearance (Two Valve Head)

Jumper pipes connect the fuel manifolds in the head with the unit injectors; one charges the unit injector, the other returns fuel to the return manifold.



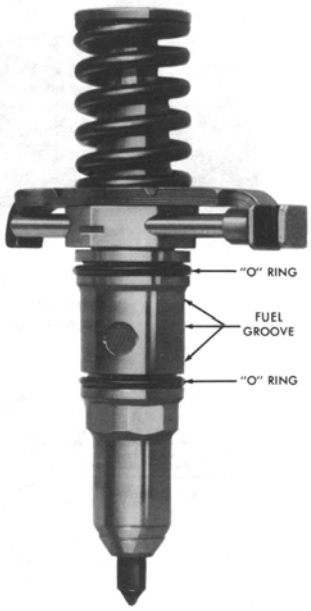


FIGURE 24.4 Two- and four-stroke diesel engine injectors (courtesy Detroit Diesel Corporation, 13400 W. Outer Drive, Detroit, Michigan 48228).

Unit injectors

A unit injector combines a complete pumping, metering element and a hydraulic injector nozzle in a single cam actuated unit. There are two types. However, the Crown valve type with NOP values of 800 psi can be considered obsolete and will not be discussed here. The Needle valve type was introduced in the early 1970's and is universal in DDC mechanical engines still in use.